Distribution	Support	PMF / PDF	CDF	Mean / $\mathbb{E}[x]$	Var.	Pars.	Interpretation / Typical Scenario
Discrete distributions							
Bernoulli(p)	$x \in \{0,1\}$	$p^x(1-p)^{1-x}$	$x<0 \Rightarrow 0,\ 0 \le x < 1 \Rightarrow 1-p,\ x \ge 1 \Rightarrow 1$	p	$p(1-p)$	p	A single "yes/no" trial: success with probability p and failure with probability $1-p$.
Binomial(n, p)	$x = 0, \dots, n$	$\binom{n}{x} p^x (1-p)^{n-x}$	$\sum_{t=0}^{x} \binom{n}{t} p^t (1-p)^{n-t}$	np	$np(1-p)$	$n,\ p$	The count of successes in n independent "yes/no" trials, each with probability p.
Geometric(p)	$x = 1, 2, \dots$	$(1-p)^{x-1} p$	$1-(1-p)^x$	$1/p$	$(1-p)/p^2$	p	The number of trials required to obtain the first success, where each trial has success probability p.
Poisson(λ)	$x = 0, 1, 2, \dots$	$e^{-\lambda} \dfrac{\lambda^x}{x!}$	$e^{-\lambda} \sum_{t=0}^{x} \dfrac{\lambda^t}{t!}$	λ	λ	λ	The number of events occurring in a fixed interval when events happen independently at a constant average rate λ.
Hypergeometric(N, K, n)	$x = \max(0, n-(N-K)), \dots, \min(K, n)$	$\dfrac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}$	$\sum_{t=0}^{x} \dfrac{\binom{K}{t}\binom{N-K}{n-t}}{\binom{N}{n}}$	$n\frac{K}{N}$	$n\frac{K}{N}\left(1-\frac{K}{N}\right)\left(\frac{N-n}{N-1}\right)$	$N,\ K,\ n$	Models the probability of obtaining exactly x successes in n draws *without replacement* from a population of N items containing K total successes. The PMF returns $P(X=x)$ — the probability that exactly x of the n drawn items are successes.
Negative Binomial(r, p)	$x = r, r+1, \dots$	$\binom{x-1}{r-1} p^r (1-p)^{x-r}$	$\sum_{t=r}^{x} \binom{t-1}{r-1} p^r (1-p)^{t-r}$	$\dfrac{r}{p}$	$\dfrac{r(1-p)}{p^2}$	$r,\ p$	The number of trials needed to achieve r successes in a sequence of independent Bernoulli trials with success probability p. Models overdispersed count data or waiting times for repeated events.
Continuous distributions							
Exponential(λ)	$x \ge 0$	$\lambda e^{-\lambda x}$	$1-e^{-\lambda x}$	$1/\lambda$	$1/\lambda^2$	λ	The waiting time until the next event in a Poisson process with rate λ. (Memoryless: elapsed time does not affect remaining wait.)
Gamma(α, θ)	$x \ge 0$	$\dfrac{x^{\alpha-1} e^{-x/\theta}}{\Gamma(\alpha)\,\theta^\alpha}$	$\dfrac{1}{\Gamma(\alpha)} \gamma\left(\alpha, \frac{x}{\theta}\right)$	$\alpha\,\theta$	$\alpha\,\theta^2$	$\alpha,\ \theta$	The waiting time until the αth event in a Poisson process of rate $1/\theta$; more generally, a model for positive, skewed data (e.g., total time for α independent exponential tasks).
Normal(μ, σ^2)	$x \in \mathbb{R}$	$\dfrac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$	$\Phi\left(\frac{x-\mu}{\sigma}\right)$	μ	σ^2	$\mu,\ \sigma$	The "bell-curve" distribution for continuous data arising from many small, independent effects. Commonly used to model measurement error or natural variation.
Beta(α, β)	$0 < x < 1$	$\dfrac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$	$I_x(\alpha,\beta)$	$\dfrac{\alpha}{\alpha+\beta}$	$\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$\alpha,\ \beta$	A flexible model for a probability or proportion on (0,1), able to take U-shaped, flat, or peaked forms. Frequently used as a prior for an unknown success probability.
Uniform[a, b]	$a \le x \le b$	$1/(b-a)$	$(x-a)/(b-a)$	$\dfrac{a+b}{2}$	$(b-a)^2/12$	$a,\ b$	All values between a and b are equally likely. Used when only bounds are known and no particular sub-interval is preferred.
Cauchy(x_0, γ)	$x \in \mathbb{R}$	$\dfrac{1}{\pi\gamma\left[1+\left(\frac{x-x_0}{\gamma}\right)^2\right]}$	$\dfrac{1}{\pi}\arctan\left(\dfrac{x-x_0}{\gamma}\right)+\dfrac{1}{2}$	undefined	undefined	$x_0,\ \gamma$	A heavy-tailed distribution without a defined mean or variance; arises in resonance behavior and as the ratio of two standard normal variables. Notable for extreme sensitivity to outliers.

Probability Basics

- **Sample space** Ω; events $A \subseteq \Omega$.
- **Axioms:** $0 \le P(A) \le 1$, $P(\Omega) = 1$, $P(\cup_i A_i) = \sum_i P(A_i)$ for disjoint A_i.
- **Complement:** $P(A^c) = 1 - P(A)$.
- **Union:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Independence:** $P(A \cap B) = P(A)\,P(B)$ $(\Leftrightarrow P(A \mid B) = P(A))$.
- **Conditional:** $P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$, $P(B) > 0$.
- **Bayes:** $P(B \mid A) = \dfrac{P(A \mid B)\,P(B)}{P(A)}$, $P(A) = \sum_i P(A \mid B_i)\,P(B_i)$.

  B_i is usually B and B^c or $P(A) = P(A \cap B) + P(A \cap B^c)$

Random Variables & Moments

- Discrete RV: PMF $P(X = x)$. Continuous RV: PDF $f(x)$, CDF $F(x) = P(X \le x)$.
- $E[X] = \begin{cases} \sum_x x\,P(X=x), & \text{(discrete)}, \\ \int_{-\infty}^{\infty} x\,f(x)\,dx, & \text{(continuous)}. \end{cases}$
- $E[g(x)] = \sum_x g(x) P(X = x)$
- **Linearity:** $E[aX + bY] = a\,E[X] + b\,E[Y]$.
- $\mathrm{Var}[X] = E[X^2] - (E[X])^2$, $sd[X] = \sqrt{\mathrm{Var}[X]}$.
- $\mathrm{Var}[aX + bY] = a^2\,\mathrm{Var}[X] + b^2\,\mathrm{Var}[Y] + 2ab\,\mathrm{Cov}(X,Y)$.
- **Tail-sum (integer $X \ge 0$):** $E[X] = \sum_{k=0}^{\infty} P(X > k)$.
- **Law of total expectation:** $E[X] = E\big[E[X \mid Y]\big]$.

Covariance & Correlation

- **Definition:** $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$, $\mathrm{Cov}(X,X) = \mathrm{Var}[X]$.
- **Linearity:**
  - $\mathrm{Cov}(X, aY + bZ) = a\,\mathrm{Cov}(X,Y) + b\,\mathrm{Cov}(X,Z)$
  - $\mathrm{Cov}(aX + bY, Z) = a\,\mathrm{Cov}(X,Z) + b\,\mathrm{Cov}(Y,Z)$
- **Symmetry:** $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$
- **Correlation coefficient:** $\rho_{XY} = \dfrac{\mathrm{Cov}(X,Y)}{\mathrm{sd}[X]\,\mathrm{sd}[Y]}$, $-1 \le \rho_{XY} \le 1$
- **Covariance matrix:** $\Sigma_{jk} = \mathrm{Cov}(X_j, X_k)$, or $C = \frac{1}{n} A^\top A$ *(symmetric, with variances on the diagonal)*

Expectation, Variance & Identity Rules

- **Linearity of Expectation:** $\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y]$
- **Variance rules:**
  - $\mathrm{Var}[aX + b] = a^2\,\mathrm{Var}[X]$
  - $\mathrm{Var}[X + Y] = \mathrm{Var}[X] + \mathrm{Var}[Y] + 2\,\mathrm{Cov}(X,Y)$
  - $\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$

Sampling Statistics

- Sample mean $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$: $E[\bar{X}] = \mu$, $sd(\bar{X}) = \frac{\sigma}{\sqrt{n}}$.
- Sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ (unbiased for σ^2).
- Student t: $T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ (normal data).

Central Limit Theorem

- For i.i.d. X_i with mean μ, var. σ^2, $\dfrac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1)$ as $n \to \infty$.
- Hence $S_n = \sum_{i=1}^{n} X_i \approx \mathcal{N}(n\mu, n\sigma^2)$ for large n. *With large enough sample size n the distribution will become approximately normal*

Inference

t-Distribution (Small Samples)

- Assume $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, σ unknown.
- Sample stats: $\bar{X}$, S^2.
- Test statistic: $T = \dfrac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.
- $(1-\alpha)$ CI for μ: $\bar{X} \pm t^*_{n-1,\alpha/2}\,\dfrac{S}{\sqrt{n}}$.
- Use when $n \lesssim 30$ and σ unknown.
- $\Phi(t) = P(t \le T)$
- P-val $= 2(1 - \Phi(t))$

Example: $n = 10$, $\bar{X} = 5.2$, $S = 1.3$, test $H_0 : \mu = 5$ vs $H_a : \mu \ne 5$

$t = \dfrac{5.2 - 5}{1.3/\sqrt{10}} \approx \dfrac{0.2}{0.411} \approx 0.487$

df $= 9$, two-tailed critical value $t^* \approx 2.262$

Since $|t| = 0.487 < 2.262$, fail to reject H_0: no significant difference.

Confidence Intervals (Normal)

- Assume $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$.
- $(1-\alpha)$ CI for μ: $\bar{X} \pm \begin{cases} t_{n-1,\alpha/2}\,\dfrac{S}{\sqrt{n}}, & \sigma \text{ unknown}, \\ z_{\alpha/2}\,\dfrac{\sigma}{\sqrt{n}}, & \sigma \text{ known (rare)}. \end{cases}$
- Interpretation: repeat sampling $\Rightarrow \approx (1-\alpha)$ of such intervals contain true μ.

Hypothesis Testing

- State the null hypothesis H_0 and the alternative hypothesis H_A. Select an appropriate test statistic T.
- Compute the p-value: $P\big(|T| \ge |T_{\text{obs}}| \mid H_0\big)$. Reject H_0 if $p \le \alpha$.
- For a one-sample t-test: $T = \dfrac{\bar{X} - \mu_0}{S/\sqrt{n}}$, which follows a t-distribution with $n-1$ degrees of freedom under H_0.

Confidence Intervals (Diff in two population proportions)

- Given:

  - Sample proportions: $\hat{p}_1$, $\hat{p}_2$
  - Sample sizes: n_1, n_2
  - Critical value (1.96 for 95% CI): z^*
- Standard Error: SE $= \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$
- Two-Proportion CI: $(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE$

Model Fitting (MLE & MoM)

Maximum Likelihood Estimation (MLE):

- Suppose $X_1, \ldots, X_n \overset{\text{iid}}{\sim} f(x \mid \theta)$.
- Likelihood: $L(\theta) = \prod_{i=1}^{n} f(x_i \mid \theta)$.
- Log-likelihood: $\ell(\theta) = \sum_{i=1}^{n} \log f(x_i \mid \theta)$.
- Solve $\nabla_\theta \ell(\theta) = 0$ (score function) for $\hat{\theta}_{\text{MLE}}$.
  - *Take the derivative with respect to each theta and set to* 0

Method of Moments (MoM):

- Let $X_1, \ldots, X_n$ have k unknown parameters, $j = 1, \ldots, k$.
- Equate $E[X^j] = \frac{1}{n}\sum_{i=1}^{n} x_i^j$ for each j.
- Example ($k = 2$): Suppose $X_i \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$. Then $E[X] = \mu$, $E[X^2] = \mu^2 + \sigma^2$. Equate to $m_1 = \frac{1}{n}\sum_{i=1}^{n} x_i$, $m_2 = \frac{1}{n}\sum_{i=1}^{n} x_i^2$, which yields $\hat{\mu}_{\text{MoM}} = m_1$, $\hat{\sigma}^2_{\text{MoM}} = m_2 - m_1^2$.

Linear Models (OLS)

- **Simple regression:** $Y = a\,X + b + \varepsilon$, $E[\varepsilon] = 0$. $\hat{a} = \dfrac{\text{Cov}(X,Y)}{\text{Var}(X)}$, $\hat{b} = \bar{Y} - \hat{a}\,\bar{X}$.
- **Multiple regression:** $Y = X\beta + \varepsilon$, $\hat{\beta} = (X^\top X)^{-1} X^\top Y$.
- Residuals $e_i = Y_i - \hat{Y}_i$; check homoskedasticity, independence, etc.

Principal Component Analysis

- Compute covariance/correlation matrix Σ of (standardized) data.
- Eigenvectors v_i of Σ are PCs; eigenvalues λ_i give variance explained.
- First k PCs capture $\left(\sum_{i=1}^{k} \lambda_i\right) / \left(\sum_{j} \lambda_j\right)$ of total variance.
- Use a scree plot (eigenvalue vs. PC index) to choose k.

Distribution Notes

- **Poisson:** $E[N] = \text{Var}[N] = \lambda$. Overdispersion if $\text{Var}(N) > \bar{N}$.
- **Exponential:** Memoryless: $P(T > s + t \mid T > s) = P(T > t)$.
- **Gamma:** Waiting time for α Poisson events. $E[X] = \alpha\theta$, $\text{Var}(X) = \alpha\theta^2$.
- **Beta:** Support $(0, 1)$; very flexible shapes for proportions.
- **Normal:** Ubiquitous via CLT. PDF: $\phi(x) = \frac{1}{\sqrt{2\pi}\,\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$.
- **Poisson Limit Theorem**: States that if n is large, and p is small $\Rightarrow Binomial(n, p) \approx Poisson(\lambda = np)$

Robust Models

- Use when errors are non-Normal (heavy tails/outliers).
- Standard OLS: $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$, minimize $\sum (y_i - x_i^\top \beta)^2$.
- Alternative: Cauchy errors: $Y_i = x_i^\top \beta + \varepsilon_i$, $\varepsilon_i \sim \text{Cauchy}(0, \gamma)$.
- Cauchy negative-log-likelihood grows more slowly $\rightarrow$ less sensitivity to large errors.

Generalized Linear Models

- **Goal:** Handle non-Normal responses and/or non-constant variance.
- **Three parts:**
  - *Random part:* Exponential-family distribution (Bernoulli, Poisson, …).
  - *Systematic part:* Linear predictor $\eta = X\beta$.
  - *Link:* $g(\mu) = \eta$, where $\mu = E[Y]$.
- **Fitting:** Maximize log-likelihood via IRLS (iteratively re-weighted least squares) or other numeric method.
- **Interpretation:** Depends on $g(\cdot)$. Coefficients on scale of η; back-transform via g^{-1} to original scale.

Logistic Regression

- Model: $Y \sim \text{Bernoulli}(p)$ (or Binomial for grouped data).
- **Logistic link:** $\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \eta = X\beta$, $p = \frac{1}{1+e^{-\eta}}$.
- **Interpretation:** β_j = change in log-odds per unit X_j; $\exp(\beta_j)$ = odds-ratio.
- **Two-point rule:** Given (x_1, p_1) and (x_2, p_2), $b = \dfrac{\text{logit}(p_2) - \text{logit}(p_1)}{x_2 - x_1}$, $a = \text{logit}(p_1) - b\,x_1$.
- Minimize with Log-Likelihood: $L(\beta) = \prod_{i=1}^{n} p_i^{Y_i} (1 - p_i)^{1 - Y_i}$

Poisson Regression

- Model: $Y \sim \text{Poisson}(\mu)$, $E[Y] = \mu$, $\text{Var}(Y) = \mu$.
- **Log link:** $\log(\mu) = \eta = X\beta$, $\mu = e^{\eta}$.
- **Interpretation:** β_j = change in log-rate per unit X_j, so e^{β_j} = multiplicative factor.
- **Check:** If $\text{Var}(Y) > E[Y]$, consider quasi-Poisson or negative-binomial.

Transformations & Diagnostics

- **Why?** Check/fix linear-model assumptions: linearity, homoskedasticity, Normal errors.
- **Residual plots:** Plot $e_i = Y_i - \hat{Y}_i$ vs.
  - Fitted values: curvature non-linearity.
  - Predictors/fitted: funnel heteroskedasticity.
- **Transformations** (log, $\sqrt{\cdot}$, Box–Cox, …):
  - Straighten relationships, stabilize variance, reduce skew.
  - Apply to Y, to X, or both; refit, re-check plots.
  - Note: coefficients now on transformed scale (e.g. $\log X$).
- **Residual SD** (s_e): typical prediction error—lower s_e means tighter fit.

Regularisation & Cross-Validation

- **Overfitting:** Model captures noise $\rightarrow$ poor generalization.
- **Cross-Validation (CV):**
  - Split data into k folds.
  - For each fold: train on $k - 1$ folds, test on the held-out fold.
  - Compute test metric (e.g. MSE) for each fold; average $\rightarrow$ CV error.
  - Compare models/penalties by CV error; CV error training error $\rightarrow$ overfitting.
- **Regularisation:** Add penalty $\lambda \times$ (penalty) to loss: obj = loss + λ penalty.
  - Ridge: $\lambda\|\beta\|_2^2$.
  - Lasso: $\lambda\|\beta\|_1$.
  - Elastic-net: combination of ℓ_1 and ℓ_2.
  - Poisson ℓ_2: for count-data penalisation.
- Choose λ by CV to minimize CV error.

Log and Prod Rules

- $\log(ab) = \log a + \log b$, $\log(a/b) = \log a - \log b$, $\log(a^c) = c \log a$.
- $\log\left(\prod_i f_i\right) = \sum_i \log f_i$, $\prod_{i=1}^{n} c = c^n$, $\prod_{i=1}^{n} c^{x_i} = c^{\sum x_i}$